Euler Method

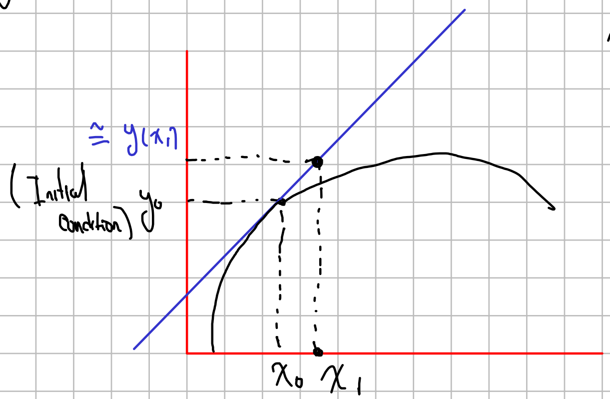
You've got a function that can't be solved by 'y'

- $\frac{dy}{dx} = f(x,y)$ — $y(x) = ??$
- $y(x_0) = y_0$

Initial condition $y(x_0) = y_0$

$$L(x) = \frac{dy}{dx}(x_0)(x - x_0) + y(x_0)$$

$$= f(x_0, y_0)\underbrace{(x - x_0)}_{h} + y_0$$



x_0, y_0

$x_1, \approx y_1 \to (x_1, y_1)$

Euler Method

(x_0, y_0)

$(x_1, y_1) \to y_1 = y(x_1) \approx f(x_0, y_0)(\overbrace{x_1 - x_0}^{h}) + y_0$

$(x_2, y_2) \to y_2 = y(x_2) \approx f(x_1, y_1)(\overbrace{x_2 - x_1}^{h}) + y_1$

$\vdots$

$(x_n, y_n) \to y_n = y(x_n) \approx f(x_{n-1}, y_{n-1}) \cdot h + y_{n-1}$